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Cybersecurity in 2041

Sanjiv Kela, Director, Net Protector Anti Virus

Twenty years ago the target audience of cyber crime were businesses because there just weren't as many users accessible to cybercriminals back then. By 2041, the world's population is expected to be 9.2 billion people, and the internet penetration is expected to be in excess of 75%, or about 7 billion people connected to a network that is as vulnerable as the least protected device on it.

Today we're already connecting to the internet with multiple devices – primarily a smartphone, but also perhaps using a laptop or desktop, a tablet, various wearable devices (smartwatches or fitness bands). Even our TVs, "smart" home devices such as lights and security systems, and app-connected appliances, all talk to one another, exchange data and authorisation information, and most likely are doing it all over a wireless network that's enabled by your home router. Offices have even more networking complexity, and have to deal with not just work devices but also all of the personal devices that people carry into the offices with them. During the pandemic, offices have had to evolve fast to allow people to be able to work from home, and give them access to data that is usually a lot easier to secure on site for a business.

The future is going to see an exponential increase in the use of technology. Because technology and cybersecurity are tightly coupled to one another, an exponential increase in technology complexity means that security firms like mine have to plan for an exponential increase in cybersecurity threats, which, honestly, is very exciting for any security researcher.

India is expected to overtake China as the most populous country soon (as early as 2022, but definitely by 2027). With the rapid progress of technology penetration across our population, and the pandemic forcing technology adoption in even the poorest sections of our society, India is also expected to be the country with the most internet users well before 2040. The cybersecurity platforms of India will be held up to the world as examples, and Net Protector is gearing up to play our role in this exciting future.

We see our role of protecting businesses, consumers, and all of the data shared between them, against cybercriminals changing and adapting as the underlying technology develops. There are five key areas in which we foresee the most changes happening, and of course, these will also result in the most exciting challenges.

1) Ransomware attacks.

There will be a rise in Ransomware Attacks. Both individuals and organisations will be targets of a Ransomware attack, and businesses will have to use solutions such as NPAV Total Security, which prevent ransomware attacks. We are continuously evolving to predict and stop these attacks.

2) Secure Remote Access.

The recent shift to remote working is causing the need to guarantee security of remote access. To solve this there are methods like a remote worker having a uni-purpose laptop which is only capable of performing a single task and won't have access to email, social media, or any public network connections at all.

3) Data Leak Prevention.

Due to the hybrid style of working from home and office, companies have been using new software, cloud services, etc., for internal processes, to host their data. With increased data online there is increased complexity and the next generation will have to be refined and will use all new systems of security.

4) Internet of Things.

A recent forecast report by the US National Intelligence Council suggests that we will have 64 billion IoT devices by 2025, and this number will rise exponentially to several trillions by 2040. Each of these devices will connect via 5G networks and will need to be secured.

5) Increased Automation.

It's not just users that need to be protected, because with increased penetration of IoT devices, there will be increased automation in systems. Whether in the form of software automation, or hardware devices such as sensors automatically accessing databases to update them, there will need to be increased monitoring of systems, and that monitoring will also be largely automated and conducted by heuristic software and AI.

We are using machine learning to help us predict attacks faster and inform of attacks before they happen. We will provide an end-to-end platform to both individual users and organizations which can solve all their security needs.